

Put plant behind the gym, with a door from inside

Full-house location plan 1:100 at A3 | Garden / south up; arrival / north down | Previous arrival-side addition declined



The retained side wall overlaps 1.05 m2 of the 4.70 x 3.00 m study rectangle; net new footprint is 13.05 m2.

REVISED LOCATION

Continues the existing gym block toward the garden, beside the snug.
No projection past the arrival front.
4.00 x 2.65 m clear plant room.
13.05 m2 added external footprint.

INTERNAL AND EXTERNAL ACCESS

Boot room -> gym -> plant room.
D21: 0.90 m opening allowance.
D22: separate 1.00 m outside opening.
Frames / clear widths unselected.
External access supports replacement.

SNUG CHANGE

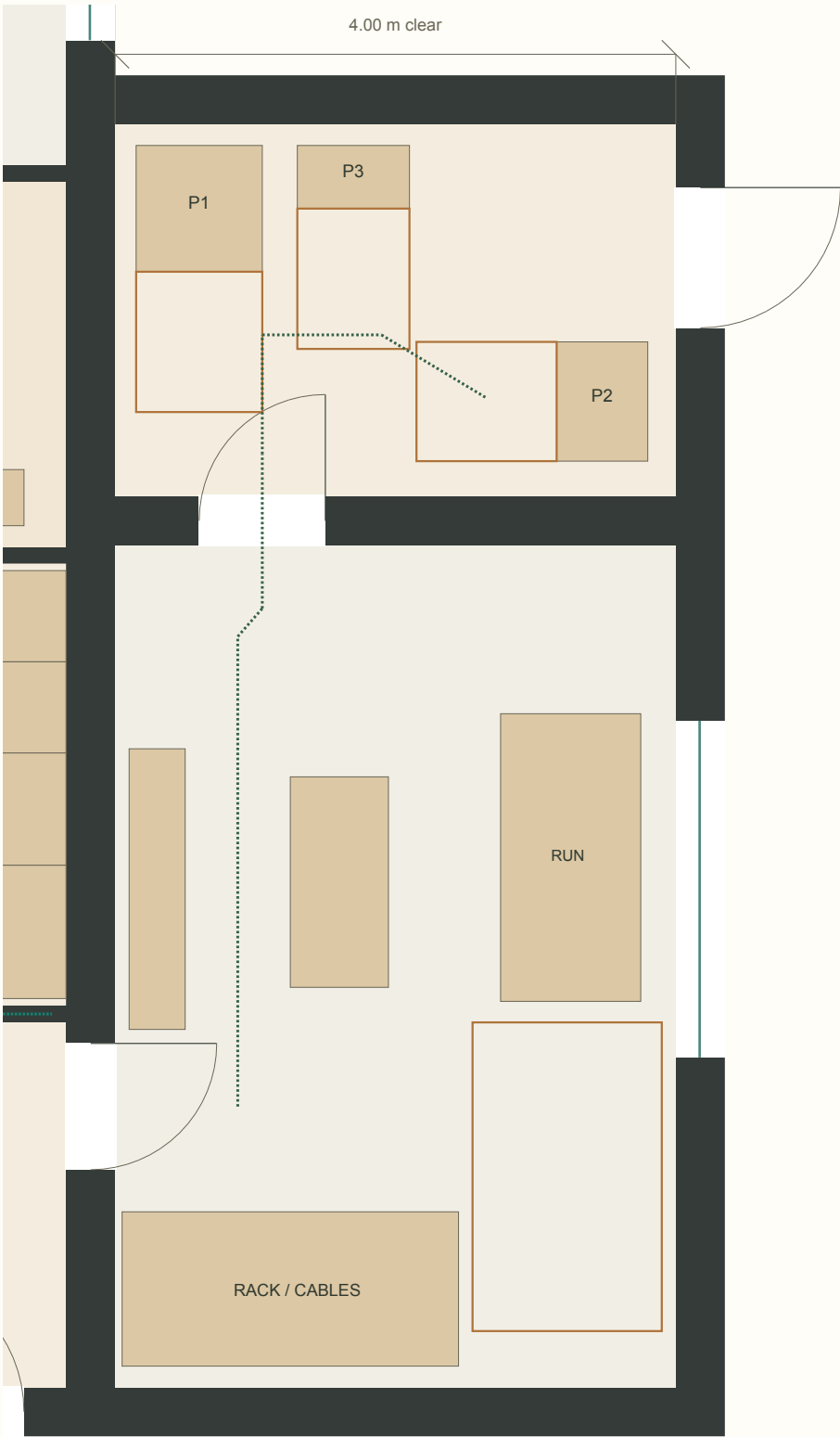
Existing side window is covered.
Replace it with a rooflight proposal.
TV, sofa and reading use stay.
A rooflight brings sky light, not a view.
Blackout and ventilation need design.

GYM CHANGE

Retains all equipment and 24 m2.
Rack and dumbbells move.
Boot-to-gym door shifts 0.60 m.
Treadmill and bench stay in place.

An interior route without reducing the gym

Gym and plant detail 1:50 at A3 | Equipment body envelopes retained | Amber = equipment service / treadmill rear reservations



PLANT FIT TEST

- P1: 0.90 x 0.90 m cylinder / connections allowance.
- P2: 0.65 x 0.85 m ventilation allowance.
- P3: 0.80 x 0.45 m hydraulic / control allowance.
- Each has a 1.00 m front floor reservation.
- Capacities, vessels, valves, ducts and drains remain unsized.

GYM REARRANGEMENT

- Rack / cables move to the arrival-end wall.
- Dumbbell storage turns onto the side wall.
- The bench and treadmill remain in their original positions.
- Retain the provisional 1.35 x 2.20 m treadmill rear zone.
- D18 moves from y=16.20 to 15.60 m to clear the rack.

WHAT HAS BEEN CHECKED

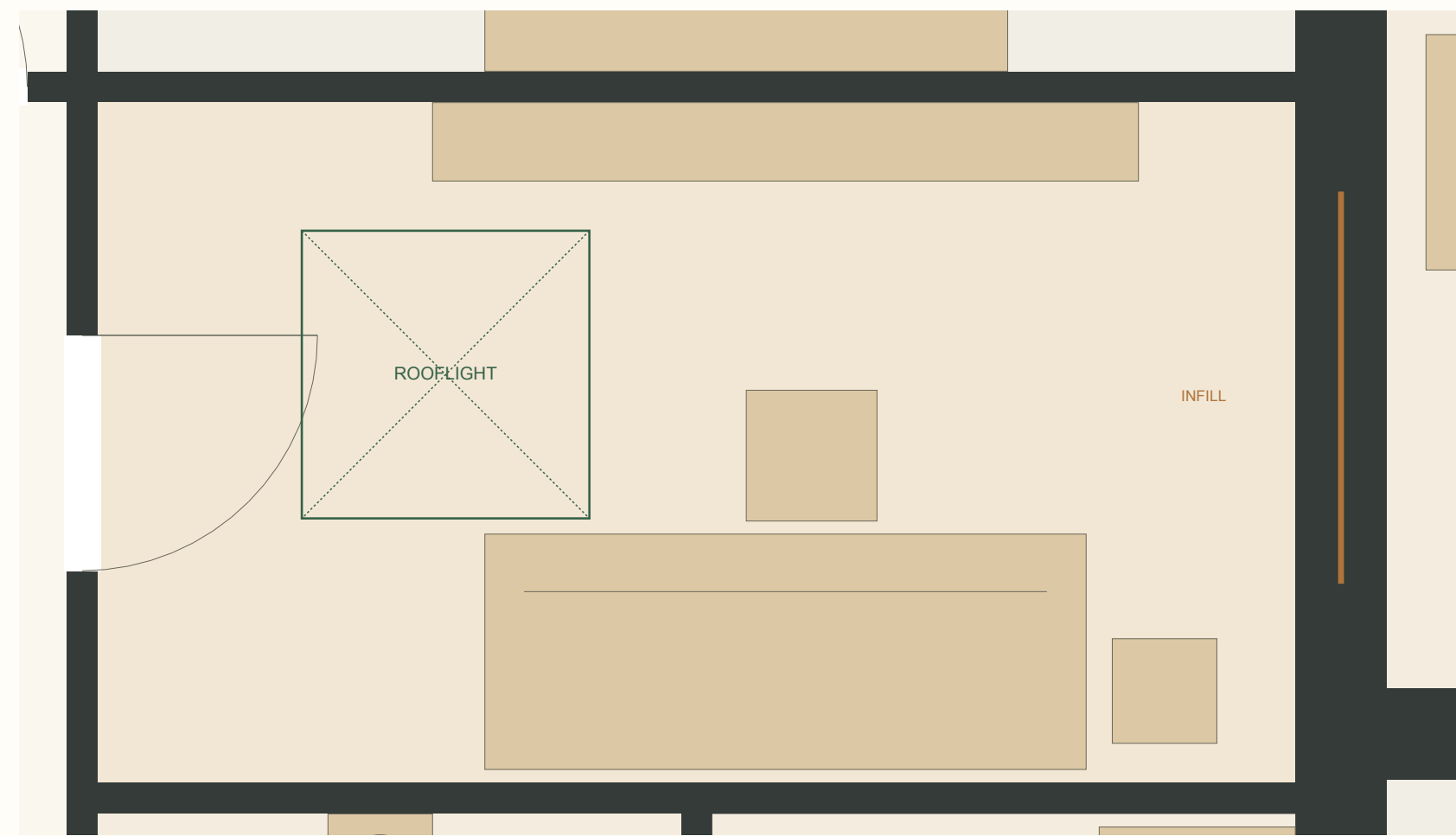
- 54 sampled house / equipment approach routes pass.
- Room doors clear the drawn equipment; internal D21 opens into plant.
- The new approach routes also clear the fully open D21 leaf.
- All existing room polygons and equipment body sizes are retained.
- Exercise movements, cable travel and barbell handling are unverified.

DETAILS STILL TO RESOLVE

- Internal door sound separation and any required fire performance.
- Acoustic mounting on the snug side; quieter equipment positioning.
- External landing, removal handling, drainage and actual boundaries.
- Roof structure and both new door lintels need coordination.

Keep the snug for films; bring daylight from above

Snug plan 1:25 at A3 | 1.10 x 1.10 m rooflight opening reservation | Existing furniture retained



THE BENEFIT

- Plant sits behind the gym, away from the arrival front.
- The snug retains its full 4.58 x 2.60 m room.
- Top light can work with a cinema / reading room.
- No plant access door is routed through the snug.

THE TRADE-OFF

- The side view and existing opening window are lost.
- Specify blackout for daytime films and assess solar gain.
- Rain noise, glare and cleaning access need review.
- Choose fixed or opening rooflight with the ventilation design.
- An opening unit needs reachable controls and weather protection.

ROOF AND ACOUSTIC COORDINATION

- Rooflight location is a spatial reservation, not a daylight calculation or a selected product.
- Check joists, trimmers, upstand, roof falls and gutter / outlet positions across the extended lower block.
- The plant-to-snug wall needs acoustic design; a rooflight does not solve plant noise.
- Keep ventilation independent of rooflight opening; reposition the general ceiling light around it.
- The plant front is only 0.25 m beyond the office window extent; assess roof-edge shading and outlook there.

Rooflight / blind options: VELUX flat-roof window accessories (linked in sources on page 6).

Separate the cooling systems by wing

Conceptual zones, not selected outdoor-unit counts or installation routes | No cross-vault cooling duct is assumed

FAMILY SIDE / COOLING GROUP A

Parents + C1 + C2 + C3

Four bedroom zones

Outdoor unit on the family-side exterior

SERVICE SIDE / COOLING GROUP B

Guest bedroom + office + snug + gym

Four room zones

Outdoor unit on the gym / service-side exterior

SHARED / GARDEN ROOM

Treat the connected living and garden space as its own load / control zone.
Connect it to one wing system only if indoor connections and simultaneous capacity allow. A separate outdoor unit or more than one emitter may be needed.
Do not count a single hallway head as cooling the closed bedrooms.

WHAT THE MULTI-SPLITS STILL NEED

Refrigerant pipes, power, condensate drainage and service access.
Wall / floor units or short local ducts matched to each room.
Room heat-gain calculations, line lengths and permitted combinations.
Acoustic siting of outdoor and indoor units; no location is fixed here.

COOLING DISTRIBUTION DIRECTION

The earlier cross-vault cooling band is no longer the assumed solution.
Daikin documents multiple indoor units per outdoor unit; the final number depends on the product and room loads. Two outdoor units are a starting idea.

VENTILATION IS A SEPARATE QUESTION

MVHR supplies fresh air and removes stale / humid air.
Multi-split cooling alone does not establish that ventilation.
Compare wing ventilation systems / local solutions against a central route; unit positions, noise, balance and shared-room supply remain open.

Hot water needs its own distribution decision

Same two simultaneous high-flow shower requirement | No cylinder size or heat-pump capacity selected

A / ONE CYLINDER AT THE SERVICE END

A separate air-to-water heat pump can serve hot water and proposed wet UFH.
That normally adds an outdoor unit beyond the cooling outdoor units.
Hot-water pipes still have to reach the family bathrooms.
Investigate insulated runs within floor / service construction, so large overhead air ducts need not cross the shared room for hot water.
Compare actual pipe lengths, waiting time and distribution heat loss.
A controlled return loop is an option to calculate, not an automatic choice.

C / A COMBINED AIR-TO-AIR + HOT-WATER PRODUCT

Ordinary air-to-air units usually do not produce domestic hot water.
Combined products exist, including Daikin Multi+ with a tank connection.
Tank size, compatible indoor units and simultaneous duties are product-specific.
The two high-flow showers and repeat-use recovery must be checked before selection.
Do not assume this arrangement also supplies the proposed wet underfloor heating.

WHAT THE DRAWING CANNOT YET ESTABLISH

Water supply pressure / flow, shower flow rates and duration, recovery and usable storage are not known.
Floor build-up, pipe zones, penetrations, insulation and service access need coordinated design.
Solar intent remains: model daylight generation and evening / night demand, with battery / grid contribution explicit.

B / LOCAL HOT-WATER STORAGE

A second store nearer the family bathrooms shortens hot delivery runs.
It adds equipment, maintenance, standing losses and connections.
It would use family-side space that was intended to become linen storage.
Heat input still needs routing or a separate source.
This conflicts with the preference to keep major plant away from bedrooms;
keep it as a comparison, not the default recommendation.

RECOMMENDED NEXT COMPARISON

Begin with A, keeping the full gym and laundry while avoiding a cooling duct crossing.
Compare C if it can satisfy the hot-water brief and the chosen heating strategy.
Use B only if the shorter hot-water runs justify extra family-side equipment.
Confirm whether winter heat stays as wet UFH or also uses the air-to-air systems.
That choice changes the plant package more than moving an equipment rectangle.

Resolve systems and the building envelope before more room detailing

Next decisions and current technical references | Preliminary concept development

1 / CONFIRM THE ARCHITECTURAL DIRECTION

- Garden-side plant with access through the gym, revised D18 and snug rooflight.
- Accept the loss of the snug side view and check plant-to-snug sound isolation.
- Keep the original plant placeholder until a complete relocation is designed.

2 / CHOOSE THE SERVICES PRINCIPLE

- Separate wing cooling; solve ventilation separately.
- Decide winter heating: retained wet UFH, air-to-air heat, or a defined combination.
- Size hot water for two simultaneous rain showers and repeat use.

3 / COORDINATE ROOF, WINDOWS AND SITE

- Actual site, orientation, boundaries, levels and outdoor-unit locations.
- Continuous lower-block roof falls, structure, rooflight and access.
- Glazing / shading / fabric assumptions for heat-loss and cooling calculations.

4 / CHECK COST AND DAILY OPERATION

- Budget the chosen footprint and system package before detailed joinery.
- Check noise, controls, maintenance access and energy use together.
- Physical room / door mock-ups remain useful for final comfort checks.

PRIMARY REFERENCES / CHECKED 13 SEPTEMBER 2026

Daikin: multi-split room connections and combined hot-water option

Energy Saving Trust: air-to-air heat pumps and usual hot-water limitation

Energy Saving Trust: air-to-water heat pumps

Zehnder: MVHR duties and ventilation system options

VELUX: light / heat / glare control for flat-roof windows

Manufacturer examples explain system types; no supplier, equipment capacity or installation layout is selected.